

DBMS Project Report Format

Project Title: [Enter Your Project Name]

Basic Information

Field	Details
Project Title	[Your project title]
Problem Statement	[Problem # and Name]
Student Name(s)	[Your name(s)]
Roll Number(s)	[Your roll number(s)]
Date	[Submission date]

1. Abstract

Write a short summary of your project (5-6 sentences). Cover what problem you solved, which technologies you used, and what you achieved.

2. Problem Statement

State the problem you are solving in 2-3 sentences.

Example: "Small businesses struggle to get loans due to lack of credit history. This project builds a micro-lending platform that connects borrowers with investors using automated credit scoring."

3. Objectives

List your main goals:

1. [Goal 1 - e.g., Design normalized database for user and transaction data]
2. [Goal 2 - e.g., Implement MongoDB for flexible product catalog]
3. [Goal 3 - e.g., Build REST APIs for frontend integration]

4. [Goal 4 - e.g., Create dashboard for analytics]

4. Technology Stack

Component	Technology Used
SQL Database	[PostgreSQL / MySQL]
NoSQL Database	MongoDB
Backend	[Node.js / Python / Java]
Frontend	[React / HTML-CSS / Angular]
Other Tools	[List any other tools used]

5. Database Design Summary

5.1 SQL Tables Created-example.

List all SQL tables you created:

1. users - Stores user account information
2. orders - Stores order transactions
3. payments - Stores payment records
4. [Your table 4]
5. [Your table 5]

5.2 MongoDB Collections Created

List all MongoDB collections:

1. products - Product catalog with flexible attributes
2. reviews - Customer reviews and ratings
3. sessions - User session data

4. [Your collection 4]

5.3 How SQL and MongoDB Work Together

Explain briefly: SQL handles [what data] and MongoDB handles [what data]. They connect through [user_id / order_id / etc.].

Example: "SQL stores user profiles, orders, and payments for ACID compliance. MongoDB stores product details, reviews, and user activity logs. The order_id links records across both databases."

6. Sample SQL Queries (5 examples)

Query 1: [Describe what it does]

Write your SQL query here

Query 2: [Describe what it does]

Write your SQL query here

Query 3: [Describe what it does]

Write your SQL query here

Query 4: [Describe what it does]

Write your SQL query here

Query 5: [Describe what it does]

Write your SQL query here

7. Sample MongoDB Queries (5 examples)

Query 1: [Describe what it does]

`db.collection.find({ ... })`

Query 2: [Describe what it does]

`db.collection.aggregate([ ... ])`

Query 3: [Describe what it does]

Query 4: [Describe what it does]

Query 5: [Describe what it does]

8. Features Implemented

List the main features of your project:

1. [Feature 1 - e.g., User registration and login with JWT]
2. [Feature 2 - e.g., Product search and filters]
3. [Feature 3 - e.g., Shopping cart and checkout]
4. [Feature 4 - e.g., Order tracking]
5. [Feature 5 - e.g., Admin dashboard]
6. [Feature 6 - e.g., Real-time analytics]

9. Challenges Faced

Challenge	How You Solved It
[Challenge 1]	[Solution]
[Challenge 2]	[Solution]
[Challenge 3]	[Solution]

10. Testing Summary

Test Type	Result
SQL queries	[Pass/Fail]
MongoDB queries	[Pass/Fail]
API endpoints	[Pass/Fail]
Frontend functionality	[Pass/Fail]

11. Screenshots / Outputs

(Describe or list your screenshots)

1. Database schema view

2. Sample SQL query output
3. Sample MongoDB query output
4. Home page / Dashboard screen
5. Any other important screen

12. Future Improvements

List 3 things you would add given more time:

1. [Improvement 1]
2. [Improvement 2]
3. [Improvement 3]

13. Conclusion

Summarizing what you learned and what your project achieves.

14. References

List any resources you used:

1. MongoDB Documentation - docs.mongodb.com
2. PostgreSQL Documentation - postgresql.org/docs
3. [Any tutorial or resource you used]
4. [Any other reference]

Appendix: Installation Steps

Step 1: Clone or download project

Step 2: Install dependencies

Step 3: Start MongoDB/other

mongod

Step 4: Start PostgreSQL

sudo service postgresql start

Step 5: Run database migrations

Step 6: Start application

start

Checklist Before Submission

- All sections filled
- SQL queries work
- MongoDB queries work/other
- Screenshots attached (if required)
- Name and roll number written
- Spell check done